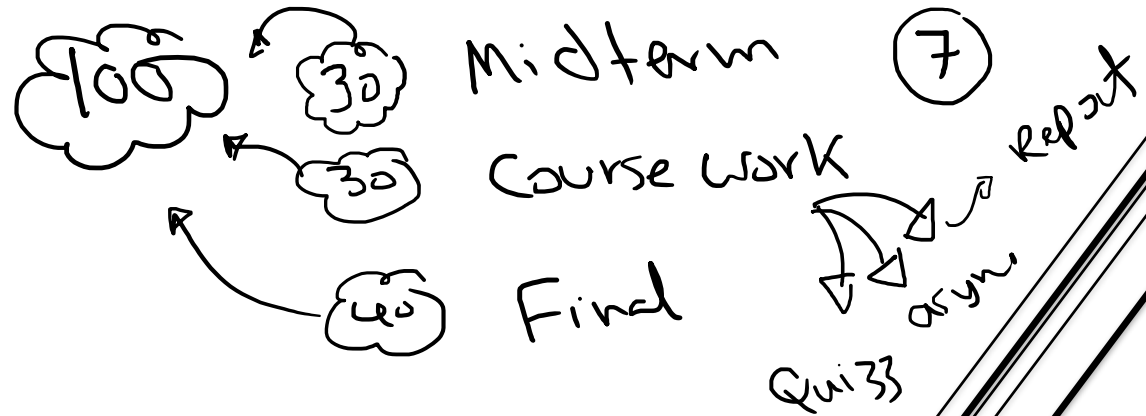




New
Mansoura
Univeristy



Analytical Geometry and Calculus 1

LMS

Lecture 1

Google Drive

MAT114



① Basics

② Line in 2D : slope, intercept
↳ first derivative

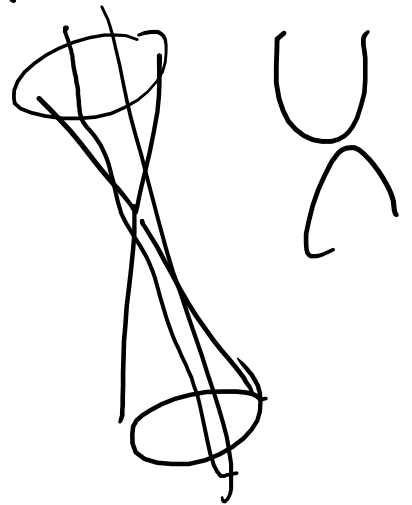
③ Functions

④ Limits & continuity

⑤ Derivatives

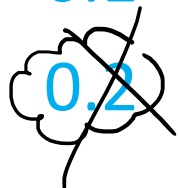
⑥ Applications of differentiation

⑦ Conic sections





0.1 POLYNOMIALS AND RATIONAL FUNCTIONS



~~0.2~~ GRAPHING CALCULATORS AND COMPUTER ALGEBRA SYSTEMS



0.3 INVERSE FUNCTIONS



0.4 TRIGONOMETRIC AND INVERSE TRIGONOMETRIC FUNCTIONS



0.5 EXPONENTIAL AND LOGARITHMIC FUNCTIONS



0.6 TRANSFORMATIONS OF FUNCTIONS



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POLYNOMIALS AND RATIONAL FUNCTIONS

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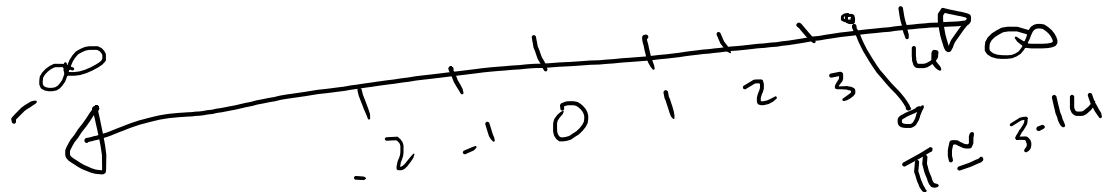
The Real Number System and Inequalities

The set of **integers** consists of the whole numbers and their additive inverses: $\{0, \pm 1, \pm 2, \pm 3, \dots\}$

A **rational number** is any number of the form p/q , where p and q are integers and $q \neq 0$. For example, $2/3$ and $-7/3$ are rational numbers.

$$\pi = 3.14 \dots \dots \dots$$
$$= \frac{22}{7}$$

Notice that every integer n is also a rational number, since we can write it as the quotient of two integers: $n = n/1$.





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POLYNOMIALS AND RATIONAL FUNCTIONS

○ The Real Number System and Inequalities

Irrational numbers are all those real numbers that cannot be written in the form p/q , where p and q are integers.

Recall that rational numbers have decimal expansions that either terminate or repeat. By contrast, irrational numbers have decimal expansions that do not repeat or terminate.

$$\begin{aligned}\sqrt{2} &= 1.41421\ 35623\ \dots, \\ \pi &= 3.14159\ 26535\ \dots \\ e &= 2.71828\ 18284\ \dots\end{aligned}$$

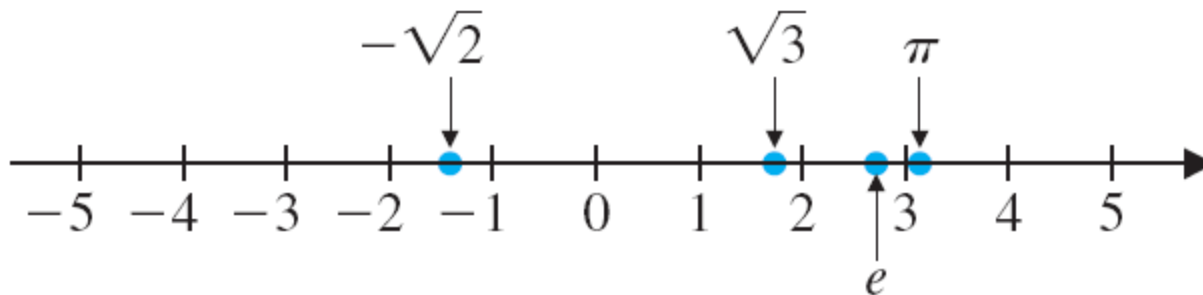


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POLYNOMIALS AND RATIONAL FUNCTIONS

○ The Real Number System and Inequalities

We picture the real numbers arranged along the number line (the **real line**). The set of real numbers is denoted by the symbol $\mathbb{R}$.





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POLYNOMIALS AND RATIONAL FUNCTIONS

$$\begin{array}{r} x+3 < 0 \\ + \\ \hline x < -3 \end{array}$$

$$\begin{array}{l} x+3 = 0 \\ x = -3 \\ \{-3\} \end{array}$$

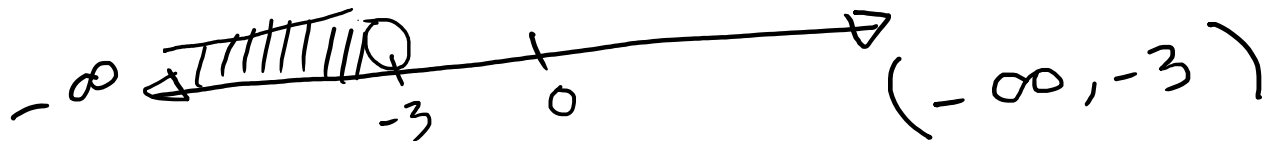
The Real Number System and Inequalities

For real numbers a and b , where $a < b$, we define the **closed interval** $[a, b]$ to be the set of numbers between a and b , including a and b (the **endpoints**).

That is, $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$.



Similarly, the **open interval** (a, b) is the set of numbers between a and b , but *not* including the endpoints a and b , that is, $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.





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POLYNOMIALS AND RATIONAL FUNCTIONS

$$a > 2 < 3 \quad c=2 \quad d=-4$$

THEOREM 1.1

If a and b are real numbers and $a < b$, then

$$4 < 7$$

- (i) For any real number c , $a + c < b + c$.
- (ii) For real numbers c and d , if $c < d$, then $a + c < b + d$.
- (iii) For any real number $c > 0$, $a \cdot c < b \cdot c$.
- (iv) For any real number $c < 0$, $a \cdot c > b \cdot c$.

$$\times 2$$

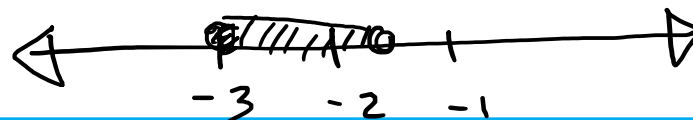
$$\begin{array}{l} 2 < 3 \\ 4 < 6 \end{array}$$

$$\left. \begin{array}{l} 2 < 3 \\ \times -2 \\ -4 > -6 \end{array} \right\}$$



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POLYNOMIALS AND RATIONAL FUNCTIONS



EXAMPLE 1.2 Solving a Two-Sided Inequality

Solve the two-sided inequality $6 < 1 - 3x \leq 10$.

$$-1 + 6 < -1 + 1 - 3x \leq -1 + 10$$

$$5 < \boxed{-3}x \leq 9$$

$$\frac{-5}{3} > x \geq \frac{-9}{3}$$

$$x \in \left[-3, -\frac{5}{3}\right)$$

**EXAMPLE 1.2** Solving a Two-Sided Inequality**Solution**

$$6 < 1 - 3x \quad \text{and} \quad 1 - 3x \leq 10$$

$$6 - 1 < (1 - 3x) - 1 \leq 10 - 1$$

$$5 < -3x \leq 9$$

$$\frac{5}{-3} > \frac{-3x}{-3} \geq \frac{9}{-3}$$

$$-\frac{5}{3} > x \geq -3 \quad -3 \leq x < -\frac{5}{3} \quad \left[-3, -\frac{5}{3}\right)$$



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POLYNOMIALS AND RATIONAL FUNCTIONS

very Important

EXAMPLE 1.3 Solving an Inequality Involving a Fraction

Solve the inequality $\frac{x-1}{x+2} \geq 0$.

$$\rightarrow x \geq 1$$

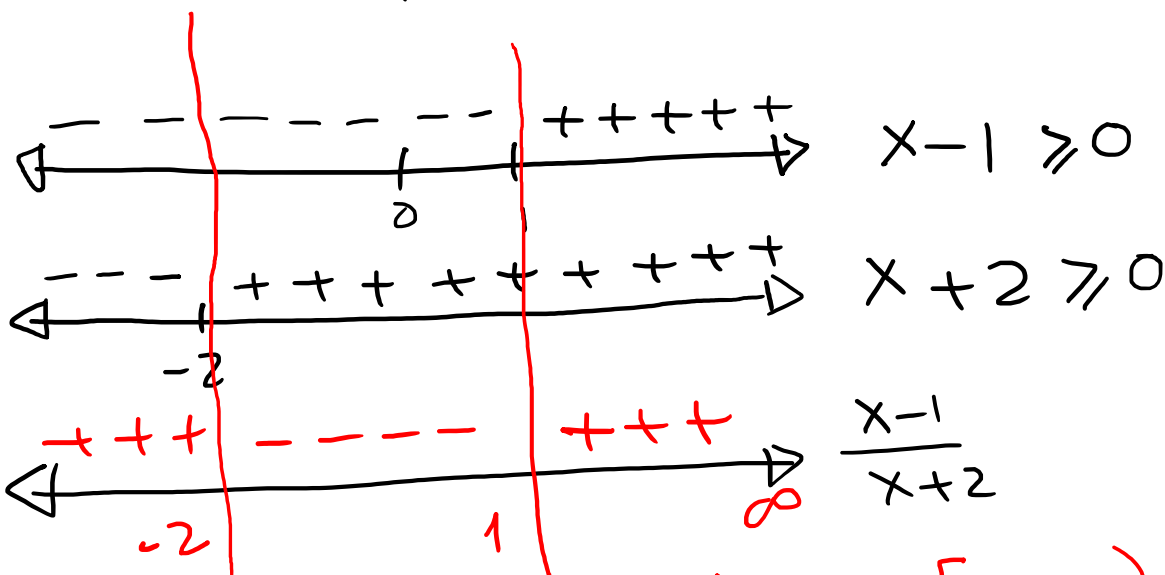
$$(x-1) \geq 0$$

$$\frac{(+)}{(+)} \geq 0$$

$$\frac{(1)}{(1)} \geq 0$$

$$\frac{(+)}{(1)} \leq 0$$

$$\frac{(1)}{(+)} \leq 0$$



$$x \in (-\infty, -2) \cup [1, \infty)$$

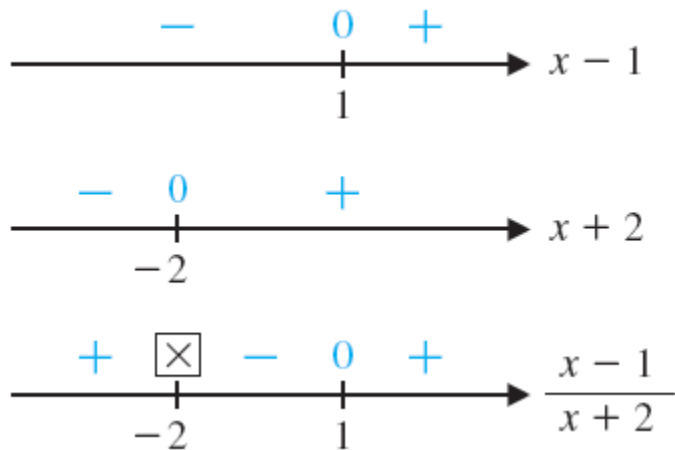


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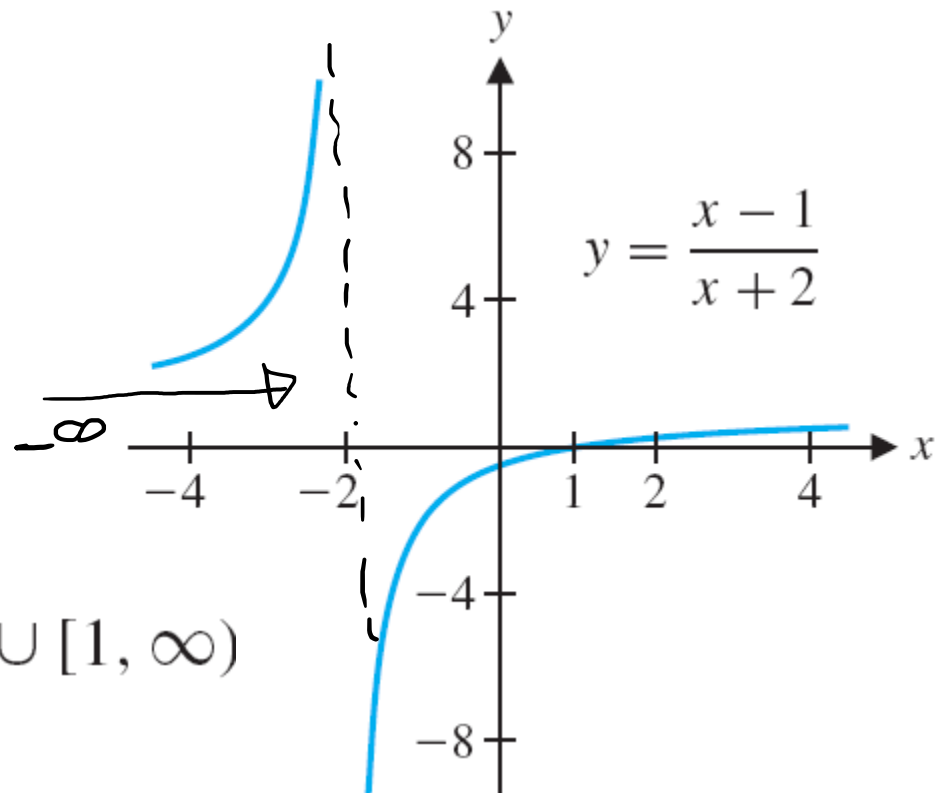
POLYNOMIALS AND RATIONAL FUNCTIONS

EXAMPLE 1.3 Solving an Inequality Involving a Fraction

Solution



$$x < -2 \text{ or } x \geq 1 \quad (-\infty, -2) \cup [1, \infty)$$





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POLYNOMIALS AND RATIONAL FUNCTIONS

DEFINITION 1.1

The **absolute value** of a real number x is

$$|x| = \begin{cases} x, & \text{if } x \geq 0. \\ -x, & \text{if } x < 0 \end{cases}$$

$$|x - 5| < 0$$

$$x \in \emptyset$$



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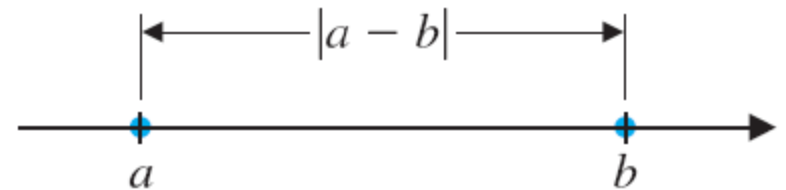
POLYNOMIALS AND RATIONAL FUNCTIONS

○ The Real Number System and Inequalities

Notice that for any real numbers a and b ,
 $|a \cdot b| = |a| \cdot |b|$, although $|a + b| \neq |a| + |b|$, in general.

However, it is always true that $|a + b| \leq |a| + |b|$. This is referred to as the **triangle inequality**.

The interpretation of $|a - b|$ as the distance between a and b is particularly useful for solving inequalities involving absolute values.





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POLYNOMIALS AND RATIONAL FUNCTIONS

EXAMPLE 1.7 Solving Inequalities

Solve the inequality $|x - 2| < 5$.

$$-5 < x - 2 < 5$$

$$-3 < x < 7$$

$$|x - 2| = \begin{cases} x - 2 & ; x - 2 \geq 0 \\ 2 - x & ; x - 2 < 0 \end{cases}$$

$$\begin{array}{l} x - 2 < 5 \\ x < 7 \end{array} \qquad \begin{array}{l} 2 - x < 5 \\ -x < 3 \\ x > -3 \end{array}$$

$$x \in (-3, 7)$$



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POLYNOMIALS AND RATIONAL FUNCTIONS

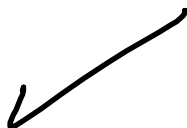
EXAMPLE 1.7 Solving Inequalities

Solution

$$-5 < x - 2 < 5$$

$$-3 < x < 7$$

$$(-3, 7)$$

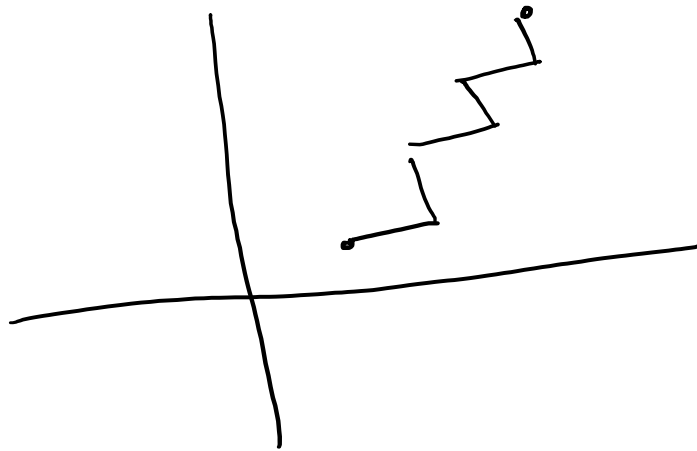




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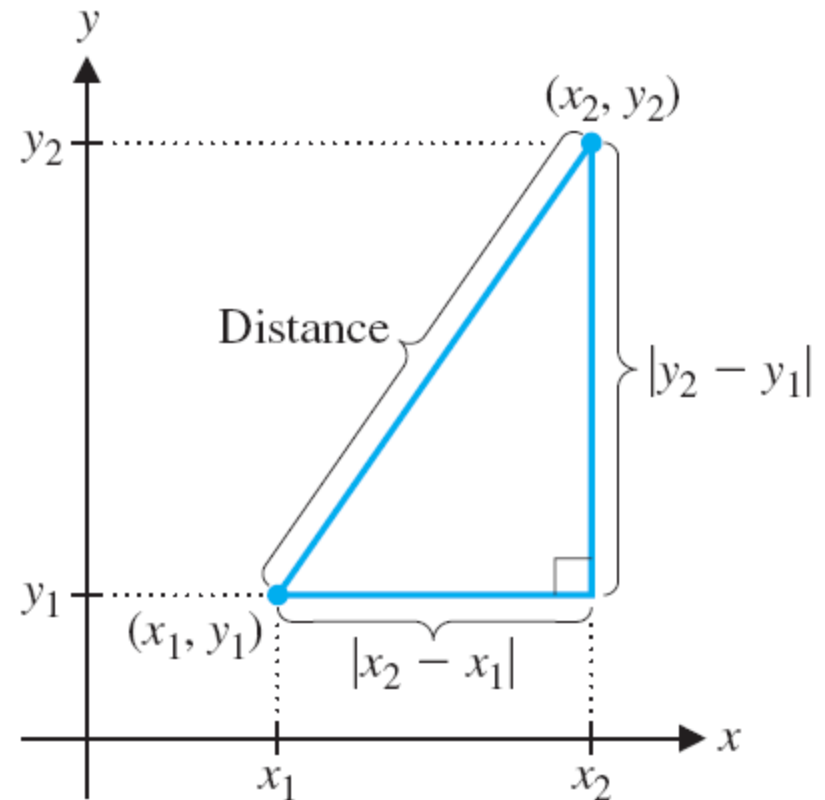
POLYNOMIALS AND RATIONAL FUNCTIONS

THEOREM 1.1



The distance between the points (x_1, y_1) and (x_2, y_2) in the Cartesian plane is given by

$$d\{(x_1, y_1), (x_2, y_2)\} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$





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POLYNOMIALS AND RATIONAL FUNCTIONS

EXAMPLE 1.8 Using the Distance Formula

Find the distance between the points $(1, 2)$ and $(3, 4)$.

$$\begin{aligned} d &= \sqrt{(1-3)^2 + (2-4)^2} \\ &= \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2} \end{aligned}$$



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POLYNOMIALS AND RATIONAL FUNCTIONS

EXAMPLE 1.8 Using the Distance Formula

Solution

$$d\{(1, 2), (3, 4)\} = \sqrt{(3 - 1)^2 + (4 - 2)^2} = \sqrt{4 + 4} = \sqrt{8}$$

